

UNIVERSITY OF PISA

COMPUTER ENGINEERING



I²C Writing Master

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Chapter 1

Introduction

The I²C is a serial communication protocol used to attach lower-speed peripheral ICs to processors and micro-controllers in short-distance and intra-board communication. As reported in Randall Hyde's book [1], before I²C bus, different components of a computer system communicated with one another using 8 to 32 data lines and some number of address signals, requiring more space and increasing the amount of noise to deal with. In 1982 [2], Philips Semiconductors (now NXP Semiconductors) developed the I²C protocol, alleviating these problems using just 2 wires: the data line (SDA) and the clock line (SCL).

1.1 Circuit description

The system is composed of one or more masters, that can write or read, and several slaves, all connected to the bus, with a bidirectional communication. Typical implementation is shown in Figure 1.1:

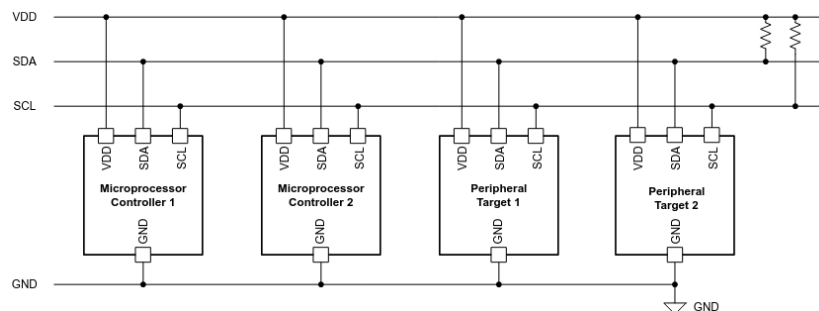


Figure 1.1: I²C implementation [2].

1.2 Possible applications

The I²C protocol is widely adopted in embedded systems, thanks to the low power consumption and space optimization, to interface microcontrollers with various peripheral ICs, such as sensors, execution elements and human interaction devices. One example is presented in [3], where the protocol is used to connect a master, the embedded computer, to all the devices installed for a smart home.

1.3 Writing Master

One possible implementation is the one reported in this documentation: the Writing Master. The device acts as the master in the I²C protocol, but can only write data; no reading operation is implemented. The design is shown in Figure 1.2:

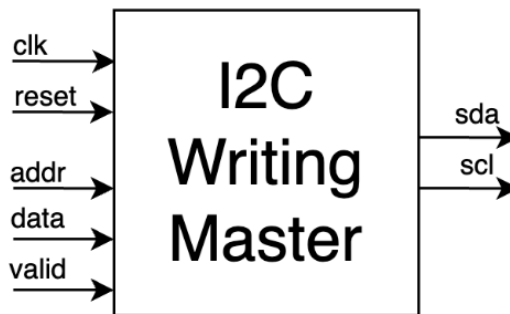


Figure 1.2: Writing Master design

The system is synchronized by a global clock (**clk**) and initialized by an active-low asynchronous **reset**. All the signals in input to the Writing Master come from a processor or a microcontroller to indicate when to communicate, using the **valid** signal, to whom, providing the 7-bit **addr** signal, and what, using the 8-bit **data** signal. After receiving the essential information, the master starts to communicate with the slave indicated in the **addr** signal using the **scl** and **sda** signals. The **sda** signal is used by the slave too during the acknowledgment phase. The clock used to communicate with the slave is 32 times slower than the system clock (**clk** signal). A possible use case of this implementation is with actuators or display drivers, for example, where the master sends a command and they simply answer with the ACK signal.

Chapter 2

Architecture

The architecture of the Writing Master is characterized by three main components:

- `REG_STATE LOGIC` (synchronous), for updating all the registers, i.e., current state, counters and data;
- `NEXT_STATE LOGIC` (combinatorial), to compute the next state;
- `OUTPUT LOGIC` (combinatorial), to generate the correct signals for the I²C protocol and serialize data.

The following section presents the block diagram to show how these components communicate and how the signals are used.

2.1 Block diagram

The Figure 2.1 shows how the three components communicate between them and which signals are used for the logic implementation:

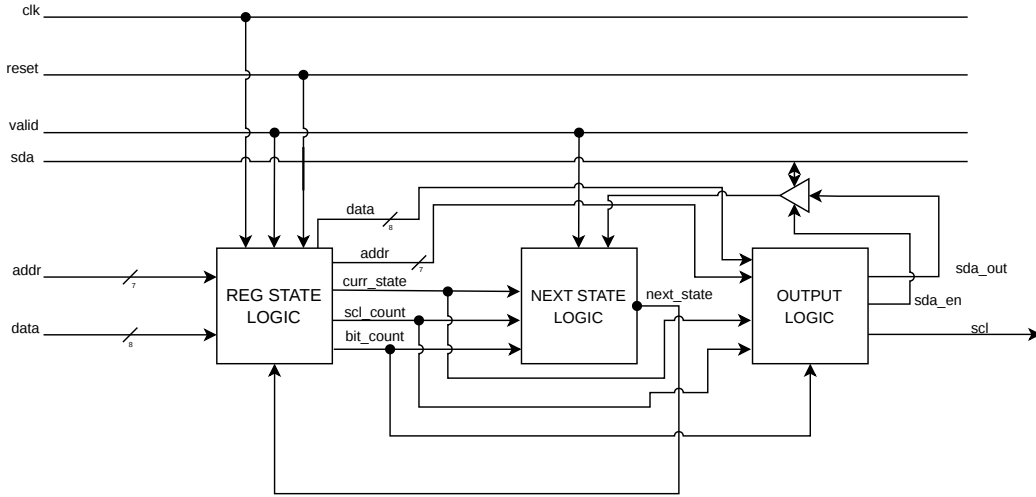


Figure 2.1: Writing Master block diagram

The **REG STATE LOGIC** is the synchronous block of the architecture. It manages the registers and handles the reset phase (**reset** signal is 0). The **next_state** signal, coming from the **NEXT STATE LOGIC**, drives the current state register, while the **addr** and **data** signals are sampled only when **valid** is 1. Regarding the counters, the **scl_count** signal is incremented by one at every clock cycle, whereas the **bit_count** signal increments every 32 clock cycles and only during transmission states. The **NEXT STATE LOGIC** is one of the two combinatorial components and implements the logic to decide the next state where the system will be at the next rising edge of the clock. It needs all the counters to know when the last bit is transmitted and the I²C cycle clock is over. Additionally, it uses the **valid** signal to start a transaction (consecutive or not) and checks **sda** signal to know if the acknowledgement signal is received. The last combinatorial component is the **OUTPUT LOGIC**, which sets the output signals to communicate using the I²C protocol. Based on the **scl_count** signal, it generates a clock 32 times slower than the system clock, while the **bit_count** signal is used to select the correct bit to transmit.

Chapter 3

VHDL

In this chapter the VHDL description of the Writing Master is presented, starting from an external view, with the entity description, to the internal architecture of the device.

3.1 Entity declaration

Listing 3.1 reports the external description of the Writing Master, as presented in Figure 1.2:

```
1 entity writing_master is
2   -- I/O ports definition
3   port (
4     clk : in std_logic;           -- system clock signal
5     reset : in std_logic;         -- reset signal
6     addr : in std_logic_vector(6 downto 0); -- slave's address
7     data : in std_logic_vector(7 downto 0); -- writing data
8     valid : in std_logic;         -- input validity signal ('1' or '0')
9     sda_in : in std_logic;        -- data line to receive ack/nack
10    sda_out : out std_logic;       -- data line to transmit serial data
11    sda_en : out std_logic;        -- enable to switch write or read on
12    sda
13    scl : out std_logic           -- device clock line
```

Listing 3.1: Writing Master Entity

The entity interfaces are presented by the `port` description, where all input and output signals are reported. As already said, the `sda` signal is a `inout` port because it is used both by the master and the slave. All the signals are based on the `std_logic` family, and not `bit` type, because they can assume other logic values rather than only 1 or 0, as the Z logic value for the `inout` signal. Every signal is represented by one bit, except for `addr` and `data` signals that are `std_logic_vector` because they are parallel buses on 7 and 8 bits respectively.

3.2 Architecture description

This section provides the internal description of the entity, including the declarative part and the statement part of the architecture.

3.2.1 Declarative part

Listing 3.2 reports all the internal signals and types used to implement the logic of the Writing Master:

```
1 end entity;
2
3 architecture structure of writing_master is
4
5     type state_t is (IDLE, START, ADDR_STATE, ACK_ADDR, DATA_STATE, ACK_DATA, STOP);
6     signal curr_state, next_state : state_t;
7     signal scl_count : unsigned(4 downto 0);           -- counter to set scl to '0'
8     or '1' (32 times slower than clk)
9     signal bit_count : unsigned(3 downto 0);           -- index of the next bit to
10    send (MSB first), max 9 (8 bit data + ACK)
11    signal addr_signal : std_logic_vector(6 downto 0);  -- set to addr when valid is
12    '1'
13    signal data_signal : std_logic_vector(7 downto 0);  -- set to data when valid is
14    '1'
```

Listing 3.2: Architecture declarative part

The type `state_t` defines all the states of the FSM. The `curr_state` and `next_state` signals are declared as `state_t` type to track the state. Regarding the counter signals, `scl_count` and `bit_count`, they are defined as `unsigned` type (5 and 4 bits respectively). Both signals are represented as a bus of wires, but they are interpreted as natural numbers allowing the use of basic arithmetic operations (`std_logic_vector` type is interpreted just as wires). Adding to this, the choice of the `unsigned` type, rather than `integer` type, is justified by the possibility to specify the exact number of bits needed (`integer` type occupies 32 bits) and to access the individual bits. The `addr_signal` and `data_signal` are used to register the values of the corresponding inputs when the `valid` signal is 1. Finally, the `scl_signal` and `sda_signal` act as internal registers to update and hold the corresponding output signals.

3.2.2 Statement part

The statement part is divided into three processes corresponding to the three blocks illustrated in Figure 2.1:

- p_STATE_REG, corresponding to the REG_STATE LOGIC block;
- p_NEXT_STATE_LOGIC, corresponding to the NEXT_STATE LOGIC block;
- p_OUTPUT_LOGIC, corresponding to the OUTPUT LOGIC block.

These three processes are illustrated in detail in the following sections.

3.2.2.1 p_STATE_REG

```
1  sda_out <= sda_out_signal;
2  sda_en <= sda_en_signal;
3
4  -- 1. State memory update
5  p_STATE_REG: process(clk, reset)
6  begin
7      if reset = '0' then
8          -- initial state
9          curr_state <= IDLE;
10         scl_count <= (others => '0');
11         bit_count <= (others => '0');
12     elsif rising_edge(clk) then
13         -- device clock is 32 times slower than clock, except for IDLE state
14         curr_state <= next_state;
15         scl_count <= scl_count + 1;
16
17         case curr_state is
18             when IDLE =>
19                 scl_count <= (others => '0');
20                 bit_count <= (others => '0');
21
22                 if valid = '1' then
23                     addr_signal <= addr;
24                     data_signal <= data;
25                 end if;
26             when ADDR_STATE | DATA_STATE =>
27                 if scl_count = 31 then
28                     bit_count <= bit_count + 1;
29                 end if;
30             when ACK_ADDR =>
31                 bit_count <= (others => '0');
32             when ACK_DATA =>
33                 bit_count <= (others => '0');
34                 if (scl_count = 31) and (valid = '1') then
35                     addr_signal <= addr;
36                     data_signal <= data;
37                 end if;
```

Listing 3.3: p_STATE_REG process

The sensitivity list of this process is composed of two signals: **reset** and **clk** (the system clock). The process triggers whenever an event occurs on one of these two signals; the internal logic evaluates the reset phase and the rising edge of the clock. During the reset phase the **curr_state** is set to the **IDLE** state and all counters are set to 0. On the rising edge of the clock the state register is updated by the **next_state** signal. Additionally, the internal counters and the data registers (**addr_signal** and **data_signal**) are updated according to the control signals.

3.2.2.2 p_NEXT_STATE_LOGIC

```
1      end if;
2      end process;
3
4      -- 2. Next state logic process
5      p_NEXT_STATE_LOGIC: process(curr_state, valid, scl_count, bit_count, sda_in)
6      begin
7          next_state <= curr_state;
8          case curr_state is
9              when IDLE =>
10                 if valid = '1' then
11                     next_state <= START;
12                 end if;
13             when START =>
14                 if scl_count = 31 then
15                     next_state <= ADDR_STATE;
16                 end if;
17             when ADDR_STATE =>
18                 if (scl_count = 31) and (bit_count = 7) then
19                     next_state <= ACK_ADDR;
20                 end if;
21             when ACK_ADDR =>
22                 -- ACK signal is sda_in low
23                 if scl_count = 31 then
24                     if sda_in = '0' then
25                         next_state <= DATA_STATE;
26                     else
27                         next_state <= STOP;
28                     end if;
29                 end if;
30             when DATA_STATE =>
31                 if (scl_count = 31) and (bit_count = 7) then
32                     next_state <= ACK_DATA;
33                 end if;
34             when ACK_DATA =>
35                 if scl_count = 31 then
36                     if sda_in = '0' then
37                         if valid = '1' then
38                             next_state <= START;
39                         else
40                             next_state <= STOP;
41                         end if;
42                     else
43                         next_state <= STOP;
44                     end if;
45                 end if;
46             when STOP =>
47                 if scl_count = 31 then
```

Listing 3.4: p_NEXT_STATE_LOGIC process

This process, corresponding to the NEXT_STATE_LOGIC block in Figure 2.1, implements the combinatorial logic to determine the next state based on the `valid` signal for the START state and on the `scl_count` signal to synchronize the clock of the I²C protocol. Furthermore, to detect the end of the sending phase and to know when the acknowledgement signal is received, the internal logic evaluates the `bit_count` and `sda` signals, respectively. In the ACK_DATA

state, a check on the `valid` signal is performed to start a new, consecutive transmission without releasing the I²C bus. To avoid inferred latches, a default assignment is provided at line 3 of Listing 3.4.

3.2.2.3 p_OUTPUT_LOGIC

```

1      end case;
2      end process;
3
4      -- 3. Output logic
5      p_OUTPUT_LOGIC: process(curr_state, scl_count, bit_count, addr_signal, data_signal)
6      begin
7          -- default ('0' with counter < 16, '1' >= 16)
8          scl_signal <= scl_count(4);
9          sda_out_signal <= '1';
10         sda_en_signal <= '1'; -- default master controls sda line
11
12         case curr_state is
13             when IDLE =>
14                 scl_signal <= '1';
15                 sda_out_signal <= '1';
16             when START =>
17                 scl_signal <= '1';
18                 if scl_count < 16 then
19                     sda_out_signal <= '1';
20                 else
21                     -- START signal with scl high and high to low transition of sda
22                     sda_out_signal <= '0';
23                 end if;
24             when ADDR_STATE =>
25                 if bit_count < 7 then
26                     -- MSB first order
27                     sda_out_signal <= addr_signal(6 - to_integer(bit_count));
28                 else
29                     -- send write (W) command
30                     sda_out_signal <= '0';
31                 end if;
32             when ACK_ADDR | ACK_DATA =>
33                 sda_en_signal <= '0'; -- slave controls sda line
34             when DATA_STATE =>
35                 sda_out_signal <= data_signal(7 - to_integer(bit_count));
36             when STOP =>
37                 -- wait a little more to avoid a new START transition
38                 if scl_count < 24 then
39                     sda_out_signal <= '0';
40                 else

```

Listing 3.5: p_OUTPUT_LOGIC process

The `p_OUTPUT_LOGIC` process, corresponding to the `OUTPUT_LOGIC` block in Figure 2.1, sets the outputs of the I²C protocol. It evaluates the `curr_state` signal, uses the `scl_count` signal to generate the I²C clock and relies on the `bit_count` signal for the correct bit to transmit on the `sda` line. As in the previous process, a default assignment is provided at lines 4 and 5 of Listing 3.5 to avoid inferred latches.

Chapter 4

Testing

The implementation described in the previous chapter is tested in four different scenarios:

- **one message**, the simplest one where the master has to send just one byte to an existing slave;
- **multiple messages**, after transmitting one message successfully, the master receives a new message to transmit to the same slave or to another, without releasing the I²C bus;
- **NACK received**, where the master receives a NACK (`sda` signal high with `scl` high) in the `ACK_ADDR` and `ACK_DATA` states;
- **asynchronous reset**, where the `reset` signal is activated (0) during the protocol.

All tests are executed with different inputs. Specifically, the **multiple messages** test is performed by transmitting to the same slave or to a different one. Furthermore, the system clock period is set to 8 ns, which corresponds to a frequency of 125 MHz. This value was specifically chosen to match the Programmable Logic (PL) clock of the ZyBo development board, as presented in the course materials.

For the details of the test bench implementation, the *tb_writing_master.vhd* file is provided.

The waveforms for the described scenarios are illustrated in the following sections.

4.1 One message scenario

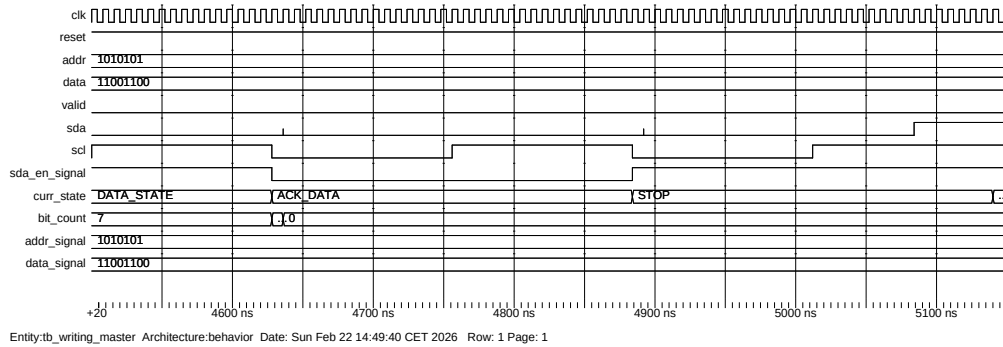


Figure 4.1: One message scenario

Figure 4.1 shows the correct behavior of the I²C Writing Master: after receiving the acknowledgement bit, the current state is set to **STOP** and then returns to **IDLE**.

4.2 Multiple messages

4.2.1 Same address

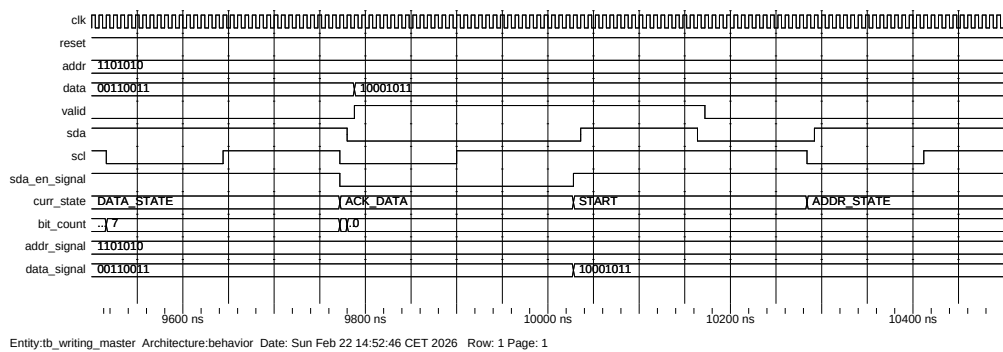


Figure 4.2: Multiple messages same address scenario

Figure 4.2 shows the correct behavior of the I²C Writing Master when a consecutive start to the same address is performed: after receiving the acknowledgement bit, the current state is set to **START** and then it initiates a new transaction.

4.2.2 Different address

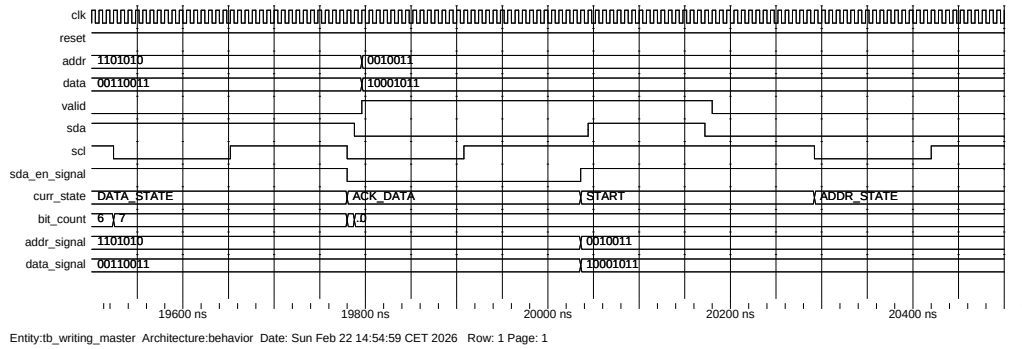


Figure 4.3: Multiple messages different address scenario

Figure 4.3 shows the correct behavior of the I²C Writing Master when a consecutive start to a different address is performed: after receiving the acknowledgement bit, a new address is read, the current state is set to **START** and then it initiates a new transaction.

4.2.3 NACK

4.2.3.1 NACK on address

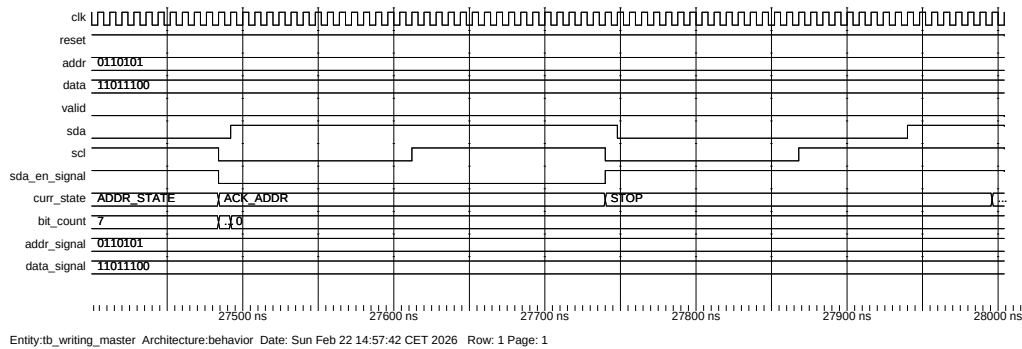


Figure 4.4: NACK received on address

Figure 4.4 shows the correct behavior of the I²C Writing Master when a NACK bit is received for the slave address: after receiving the NACK, the current state is set to **STOP**.

4.2.3.2 NACK on data

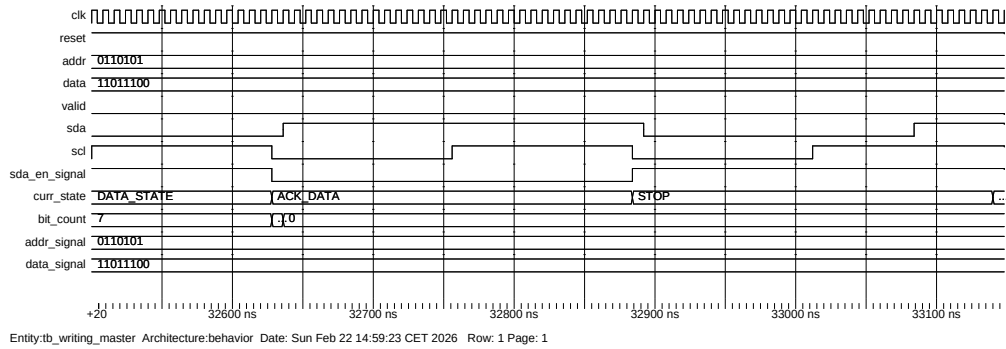


Figure 4.5: NACK received on data

Figure 4.5 shows the correct behavior of the I²C Writing Master when a NACK bit is received during data transmission: after correctly detecting the NACK (**sda** is 1), the current state is set to **STOP**.

Chapter 5

Synthesis and Implementation

The last step of this project is performed by Vivado. After the description and the testing phase have been completed, Vivado provided the synthesis and the implementation on FPGA board. The out-of-context mode has been set in the implementation phase due to the absence of physical board. In this chapter all the steps are analyzed, from the synthesis to the implementation.

5.1 Synthesis

5.1.1 Warning messages

Before illustrating the synthesis analysis, a brief comment on the warning messages encountered is reported in this section.

- **[Synth 8-5799] Converted tricell instance 'i_14' to logic**, a critical warning during the synthesis analysis due to the use of the 'Z' state in the I²C protocol. Because of the *out-of-context* mode, Vivado converts the tri-state port into combinatorial logic to simulate the correct behavior, as tri-state ports exist only on the I/O pins of the FPGA.

5.1.2 Timing analysis

After defining the clock constraint with a target frequency of 125 MHz (corresponding to a period $T_{clk} = 8$ ns), the synthesis was performed. Figure 5.1 reports the post-synthesis timing summary:

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 4.450 ns	Worst Hold Slack (WHS): 0.254 ns	Worst Pulse Width Slack (WPWS): 3.500 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 50	Total Number of Endpoints: 50	Total Number of Endpoints: 47

All user specified timing constraints are met.

Figure 5.1: Post-synthesis timing summary

As shown in Figure 5.1, the design meets the timing requirements with a *Worst Negative Slack* of **+4.450 ns**. This means that the slowest signal arrives 4.450 ns **before** the *Required time*. In Figure 5.2 is presented this critical path:

Name	Slack	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock
Path 1	4.450	4	5	7	UUT/sc1_count_reg[0]/C	UUT/FSM_seque...tate_reg[0]/D	3.528	1.163	2.365	8.0	PL_clk_125	PL_clk_125
Path 2	4.450	4	5	7	UUT/sc1_count_reg[0]/C	UUT/FSM_seque...tate_reg[1]/D	3.528	1.163	2.365	8.0	PL_clk_125	PL_clk_125
Path 3	4.450	4	5	7	UUT/sc1_count_reg[0]/C	UUT/FSM_seque...tate_reg[2]/D	3.528	1.163	2.365	8.0	PL_clk_125	PL_clk_125

Figure 5.2: Post-synthesis critical path

The most critical path comes from the `sc1_count_reg`, which is consistent with its role as the primary timing reference for the FSM. In every clock cycle the `sc1_count` is incremented, then compared to 31 and finally used to determine the next state and to update the `FSM_sequential_curr_state_reg`.

Figure 5.1 provided the data (WNS) to compute the maximum clock frequency for the Writing Master implementation. Firstly, the critical path delay (T_{dp}) is computed as follows:

$$T_{dp} = T_{clk} - \text{WNS} = 8 \text{ ns} - 4.450 \text{ ns} = 3.55 \text{ ns} \quad (5.1)$$

Finally, obtaining this period, the maximum clock frequency is calculated:

$$F_{max} = \frac{1}{T_{dp}} = \frac{1}{3.55 \text{ ns}} \approx 281 \text{ MHz} \quad (5.2)$$

Given these results, the clock frequency of the I²C protocol, which is 32 times slower than the clock system, ranges from 3.9 MHz (at a nominal 125 MHz system clock) up to a theoretical maximum of ≈ 8.7 MHz (F_{max}). The values obtained are exceptionally high and were selected to accelerate

testing and simulation analysis. As reported by [2], the I²C protocol typically operates at 100 kbps in *Standard-mode*, while the fastest operational modes (such as the *Ultra-Fast mode* up to 5 Mbps) omit key protocol features and operate in a write-only capacity.

5.1.3 Resource utilization

After the synthesis process, Vivado generates a report of the hardware utilization in terms of LUTs (combinational logic) and Registers (sequential logic). Table 5.1.3 shows the post-synthesis utilization:

Resource	Estimation	Available	Utilization %
LUT	26	17600	0.15
FF	47	35200	0.13

Table 5.1: Post-synthesis resource utilization

These post-synthesis results suggest an efficient and very low hardware utilization ($< 1\%$) for both combinatorial and sequential logic.

5.1.4 Power consumption

The post-synthesis power utilization is reported by Vivado in Figure 5.3:

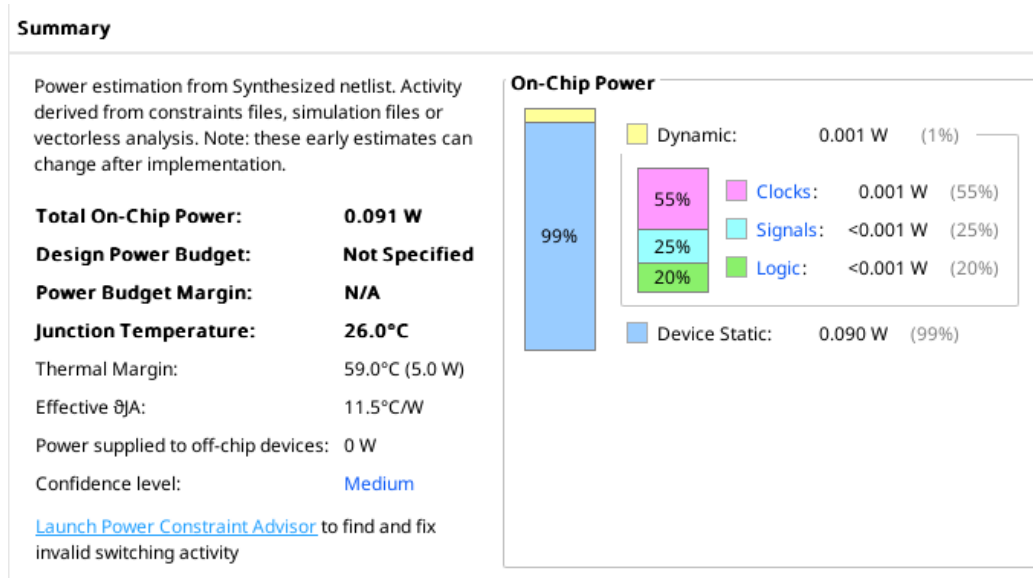


Figure 5.3: Post-synthesis power consumption

Consistent with the results of Table 5.1.3, 99% of *Total On-Chip Power* is static, while the dynamic part is 1%. This means that most of the power is consumed to keep the device powered on, whereas a negligible part is consumed by the logic implementation.

5.2 Implementation

5.2.1 Warning messages

Before illustrating the implementation analysis, a brief comment on the warning messages encountered is reported in this section.

The implementation process terminates with success, but it reports 21 warning messages. All these warnings are caused by the *out-of-context* mode, which purposely omits all the physical I/O pin constraints. However, this doesn't affect the validity of the logic implementation.

5.2.2 Timing analysis

Similar to Section 5.1.2, after the implementation process terminated successfully, the *Design Timing Summary* is illustrated in Figure 5.4:

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 3.728 ns	Worst Hold Slack (WHS): 0.157 ns	Worst Pulse Width Slack (WPWS): 3.500 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 50	Total Number of Endpoints: 50	Total Number of Endpoints: 47

All user specified timing constraints are met.

Figure 5.4: Post-implementation timing summary

These results validate and improve the post-synthesis timing estimation. The *Worst Negative Slack* is +**3.728 ns**, that is approximately the **6%** less than the synthesis expectation. The critical path is approximately the same, but with different delays, as expected:

Name	Slack	Levels	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Uncertainty
Path 1	3.728	4	7	UUT/scl_count_reg[0]C	UUT/FSM_seque...tate_reg[1]D	4.217	0.952	3.265	8.0	PL_clk_125	PL_clk_125		0.035

Figure 5.5: Post-implementation critical path

This decreased WNS indicates a lower realistic maximum frequency. In particular:

$$T_{dp} = T_{clk} - \text{WNS} = 8 \text{ ns} - 3.728 \text{ ns} = 4.272 \text{ ns} \quad (5.3)$$

$$F_{max} = \frac{1}{T_{dp}} = \frac{1}{4.272 \text{ ns}} \approx 234 \text{ MHz} \quad (5.4)$$

In conclusion, the maximum clock frequency of the I²C protocol, which is 32 times slower than the clock system, will be $\approx 7 \text{ MHz}$ (F_{max}). As for the post-synthesis analysis, this value far exceeds the standard I²C protocol speed.

5.2.3 Resource utilization

As for the synthesis analysis, Vivado generates a report of the hardware utilization in terms of LUTs (combinational logic) and Registers (sequential logic). Table 5.2.3 shows the post-implementation utilization:

Resource	Estimation	Available	Utilization %
LUT	26	17600	0.15
FF	47	35200	0.13

Table 5.2: Post-implementation resource utilization

These results confirm the utilization obtained in the post-synthesis analysis.

5.2.4 Power consumption

The post-implementation power utilization is reported by Vivado in Figure 5.6:

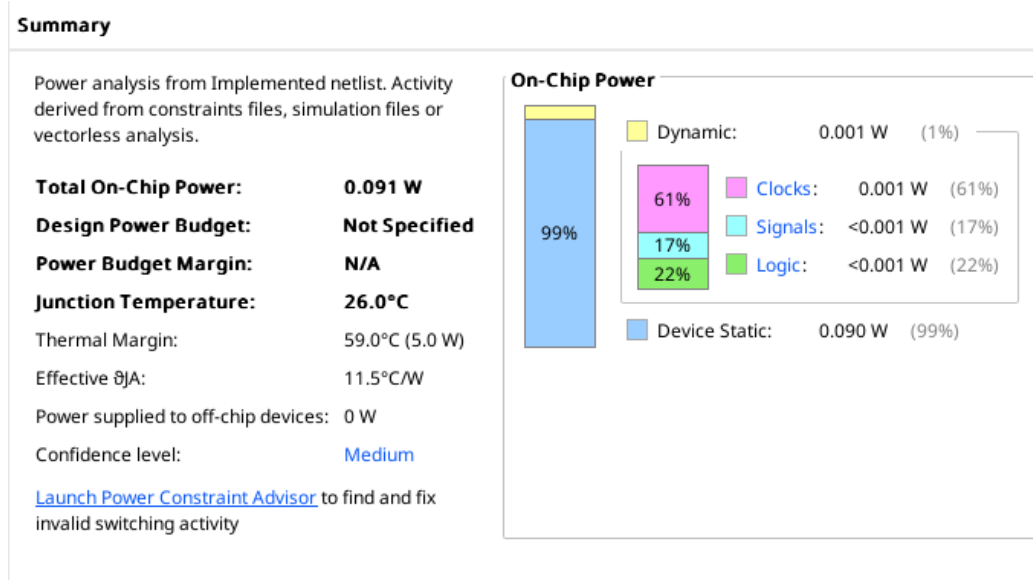


Figure 5.6: Post-implementation power consumption

The total dynamic power remains strictly minimal at 0.001 W, consistent with the synthesis estimates. However, the internal distribution of this dynamic power highlights the effects of the physical implementation. The power dissipated by *Signals* significantly dropped from 25% to 17%. This confirms the LUTs compacted placement, reducing the distance and the signal power. Consequently, the proportional weight of the *Logic* and *Clocks* slightly increased to 22% and 61% respectively. Overall, the total on-chip power is still dominated by the 0.090 W device static power, demonstrating the extreme power efficiency of the designed I²C master module.

Chapter 6

Conclusions

In conclusion, the I2C Writing Master satisfies the requirements provided and works effectively. The results provided by the test bench confirm the correct behavior of the VHDL code and the implementation shows an efficient device, with low hardware utilization and excellent power efficiency. Finally, from the computations, the theoretical clock frequency can be up to 223 MHz.

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